

FUNDX

FINS3645 FinTech Project 2026 - Part B

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Live app <https://z5544937-fundx.streamlit.app>

Public repository https://github.com/1Tehan/z5544937_projectB

Main narrative word count: 4,531 (excluding references and appendix)

Executive Result

PRODUCT DECISION

Begin with Combined Risk Parity if you are a balanced investor. This strategy does not produce the best raw returns; however, it produces the highest combined fund Sharpe of 0.862. The worst drawdown is less than that of Combined Max Sharpe by material amounts (-19.5% vs. -22.7%) and this strategy also provides the least amount of average monthly turnover at 3.7%.

- The Sharpe Ratio for the Equity Equal Weight Portfolio ranked first among all optimized equity portfolios in the OOS (Out-of-Sample) period, with a Sharpe Ratio of .784.
- Only when sizing controlled the risk did adding Crypto improve the results. The Combined Equal Weight portfolio had a lower Sharpe Ratio (.701) than Equity Equal Weight (.784). However, Combined Risk Parity performed better than Equity Risk Parity, with Sharpe Ratios of .862 and .682 respectively.
- Fin-VADER improved sentiment measurement, but this did not improve investment performance. Neutral sentiment dropped from 49.6 percent to 45.2 percent. However, the Fin-VADER Tilted portfolio produced a Sharpe Ratio of .342, below the .380 Sharpe Ratio of the Original Portfolio.
- I also used a 21-day moving-block bootstrap to avoid overclaiming the results. Since each selected fund's 90 percent Sharpe interval included zero, I also considered Drawdown and Turnover when determining the strongest overall portfolio.

1. The Funds and the Backtest Design

Fund X was established by Part A so that investors could have broad market exposure as well as a specific investment decision rule, all of which were provided without having to select each and every equity position on their own and/or go through an opaque allocation process. Part B took this and made it into an investable fund shelf. Each Fund X portfolio will detail how the portfolios are constructed, provide a Fact Sheet outlining out-of-sample historical performance, list the current holdings within the portfolio, and provide a space for comparison/combination of different funds. The Allocation Builder also outlines an illustrative 0.75% annual management fee.

Method	Role in FundX	Main estimation exposure
Equal Weight	Transparent 1/N benchmark	None beyond the investable universe
Minimum Variance	Defensive / lowest-risk option	Covariance estimates
Maximum Sharpe	Growth-oriented optimiser	Expected returns + covariance
Risk Parity (ERC; Spinu, 2013)	Balanced risk-budgeting option	Risk/covariance, no return forecast
Hierarchical Risk Parity (López de Prado, 2016)	Robust clustering-based alternative	Dependence structure, no matrix inversion

All portfolios were tested using walk-forward backtesting at each monthly rebalance (the first trading day of the month) using only prior data to set the weight. Each fund's time frame was 252-trading-days (equity/combined) and 365-calendar days (crypto). The equity/combined funds began on Jan. 04, 2021; crypto funds began on Jan. 01, 2021. Sharpe Ratios used a zero risk-free rate. Annualized returns are computed using 252 for equity/combined funds and 365 for crypto. For all funds, the out-of-sample (OOS) period ended Dec. 29, 2023 for equity/combined funds and Dec. 31, 2023 for crypto. All funds are long-only and fully-invested. Both Minimum Variance and Maximum Sharpe include a 20% per-asset limit on equity/combined and 40% on crypto. Equal-weighted portfolios inherently stay under these limits. I did not impose post-hoc caps on either ERC or HRP even though realized OOS portfolio weights for both were lower than those limits. In addition to the per-asset caps, Minimum Variance, Maximum Sharpe and ERC also shrink their estimated covariance matrices by 10% toward their diagonals for numerical reasons. There is a sentiment bias applied through limiting the sector multiplier to be between .60 and 1.40.

As described above, during the periods between rebalancing dates, I allow the weight of each security to drift according to the realized return of that security. Therefore, I measure turnover versus the portfolio that will reach the next rebalancing date. I charge 10 basis points $\times$ turnover on each rebalancing date and report both the gross and net-of-cost Sharpe ratio. I separate reporting into equity, crypto and combined funds because they provide different answers to my research question. Specifically, equity funds assess whether the allocation method provides better performance relative to a large-cap universe. Crypto funds isolate a higher-risk asset class on its 7-day calendar. Combined funds evaluate whether including crypto in a portfolio adds value through diversifying away equity-specific risks or if it simply competes with equity for total portfolio risk.

I prefer comparing methods within each type of fund before comparing similar methods across types of funds. If I had done otherwise, the strong cryptocurrency performance in 2021 would have dominated the results of comparison. Both the caps on Minimum Variance and Maximum Sharpe serve as a protective mechanism to prevent the models from producing high concentrations due to volatile estimation errors. ERC and HRP were not given post-hoc caps. As such, the backtests remain transparent and can be easily understood/reproduced by a retail investor.

BACKTEST GUARDRAIL

For verification, I also ran a falsification check: planting a +60% return on a rebalance date did not change that date's target weights. Separately, I recomputed Combined Risk Parity from fund_returns.csv and matched the saved annualised return, volatility, Sharpe, maximum drawdown and growth-of-\$1 metrics.

2. Out-of-Sample Results and Fund Fact Sheets

The most useful way to read the leaderboard is as an investment decision - which portfolio(s) to invest in - rather than a competition to get the highest return. Combined Maximum Sharpe has the largest combined-family wealth (\$1.61 from \$1), but Combined Risk Parity is the strongest balanced portfolio as it produces the best risk-adjusted return while having significantly less trading activity. All 17 portfolios are listed in Table 1. The growth, family comparisons, drawdowns, weight changes and cross-method Sharpe rankings can be found in Figures 1 through 5.

Selected fund	Ann. ret.	Ann. vol.	Sharpe	Max DD	Growth \$1	Turnover
Combined Risk Parity	14.0%	16.2%	0.862	-19.5%	\$1.48	3.7%
Combined Max Sharpe	17.3%	23.2%	0.744	-22.7%	\$1.61	28.5%
Combined Min Variance	5.5%	12.6%	0.434	-15.5%	\$1.17	13.0%
Equity Equal Weight	12.6%	16.1%	0.784	-20.2%	\$1.43	2.6%
Crypto Min Variance	59.8%	71.6%	0.836	-72.0%	\$4.08	10.2%
Crypto Max Sharpe	-3.0%	76.0%	-0.040	-85.6%	\$0.91	21.8%

ERC utilizes a log barrier formulation introduced by Spinu (2013) and the use of equal risk contributions to ensure that the far more volatile crypto assets do not receive the same capital weighting as equities solely due to being in the same investment universe. A good example of this is the Equal Weight portfolios. Equity Equal Weight provided a 0.784 Sharpe Ratio while Combined Equal Weight provided a 0.701 Sharpe Ratio, even though annualized return rose from 12.6% to 15.1%. Similarly, volatility rose from 16.1% to 21.6% because ten crypto assets were given the same nominal weighting as fifty equities. On the other hand, Combined Risk Parity performed better than Equity Risk Parity with Sharpe Ratios of 0.862 and 0.682. Therefore, the issue isn't whether or not to include crypto in a portfolio; however, the question revolves around how well the sizing rule limits crypto volatility from overwhelming the overall portfolio.

Benchmark results within the equity family contain the most impressive performance result. Equity Equal Weight outperformed each and every optimized equity-only portfolio on Sharpe ratio: .784 versus .682 for

Risk Parity, .611 for HRP, .436 for Minimum Variance and .380 for Maximum Sharpe. This is consistent with DeMiguel et al. (2009) whereby out-of-sample estimation error can remove the theoretical benefits associated with optimization. Additionally, since Maximum Sharpe estimates both expected returns and covariance, whereas Equal Weight estimates none and also exhibits the lowest turnover of all at 2.6%, this clearly demonstrates that adding complexity does not always yield a better product.

Minimum Variance exhibited approximately what its name suggests, with about 12.6% annualized volatility and the smallest draw-downs among the equity and combined families (-15.3% and -15.5%) while exhibiting an annualized return of just 5.5%. HRP followed López de Prado (2016), grouping dependence relationships together and allocating recursively rather than directly inverting the covariance matrix. Combined HRP exhibited 14.0% volatility, a -17.9% draw-down and a 0.701 Sharpe Ratio. Its contribution comes mainly from an alternative portfolio construction method rather than clearly improved performance.

The crypto-only family is arguably the most evident demonstration of why utilizing return alone to evaluate a fund can lead to poor decision-making. Crypto Minimum Variance exhibited an annual return of 59.8% and grew \$1 to \$4.08, yet still exhibited a -72.0% maximum draw-down. Crypto Equal Weight reached a -81.6% draw-down, Crypto HRP achieved a -78.1% draw-down, and Crypto Maximum Sharpe exhibited a -3.0% annual return, -0.040 gross Sharpe, -0.043 net Sharpe, \$0.91 ending wealth and an -85.6% draw-down. These funds continue to exist, albeit they should probably be viewed strictly as high-risk crypto exposure rather than as the default diversified FundX product.

Similarly, Combined Maximum Sharpe illustrates the trade-offs involved in selecting between return and concentration, and ultimately implementation. It yielded an annualized return of 17.3% and \$1.61 ending wealth, yet its volatility was 23.2%, its Sharpe ratio was 0.744, and its maximum draw-down was -22.7%. It peaked at approximately \$1.73 in April 2022 and subsequently suffered through its deepest draw-down without recovering completely from that event prior to the conclusion of the sample period. This is particularly relevant for retail investors who enter positions close to a historical peak, which can produce a very different investment experience.

Therefore, my investigation provides evidence supporting FundX's utilization of a "shelf" model instead of one universally best performer. Minimum Variance is suited for those seeking to prioritize controlling losses, while Equal Weight continues to represent a clear and competitive benchmark, Maximum Sharpe represents a higher return but more concentrated option, and Risk Parity represents a balanced default choice in this sample set. I would not recommend promoting crypto-only funds based upon their high average annualized returns due to the large draw-downs exhibited by these funds ranging from roughly -72% to -86%. The results from the combined-family provide a strong basis for FundX's strategy of creating a diversified product.

INTERPRETATION RULE

Figure 1 and Figure 5 deliberately tell different stories: Maximum Sharpe finishes with the most money in the combined family, while Risk Parity has the better Sharpe. FundX should not collapse those two questions into a single "best fund" label.

3. The Standalone Sentiment Index

The Part B ETL logic was identical to Part A's, and followed all of the same data rules. Using ticker, date and title; we removed duplicate headlines and reduced our number of headlines by 2847 resulting in a total of 146,836 unique headlines. We converted the date field into a UTC-aware format and assigned each headline to the appropriate trading day (if available) or the next trading day. We decided to keep the raw headline text intact due to the fact that VADER measures sentiment based on punctuation, capitalization, surrounding words etc.

Each headline received a compound score which we first averaged by ticker-day and then by sector-day. We used equal weights for each ticker with news because if one ticker had more news than another we would not want to give that ticker more weight. Since zero is an actual VADER neutral score, and no headline means there is no observation, we did not replace missing ticker-days with zero. Replacing those values with zero

could potentially make sectors with less news look more stable than they really are. We also lagged sentiment by one trading day so information from day t cannot be used until day $t+1$.

As shown in Figure 6 the average of the 21-day sector indices were mostly above zero. On average, using Plain VADER the headlines have a +0.106 rating, where 37.4% are positive, 13.0% negative, and 49.6% neutral. As seen during the March 2020 COVID-19 shock when OXY fell 52% the index was affected by the event but did not become strongly negative. This suggests that the index reflects more than just market stress, including company announcements, publisher selection, and differences in what gets reported. However, Utilities remained higher than most other sectors; however since news coverage can vary depending on the sector I consider this to be more descriptive.

Therefore, I will treat the index as a relative measure of news tone rather than a direct measure of investor sentiment. It only uses headlines, and general-purpose VADER still produces a large neutral share. Part A already predicted that roughly half of the headlines would be neutral so the actual 49.6% supports limitations identified before results were known for Part B. Also, the positive average score does not mean investors were consistently optimistic. Due to these limitations we developed both the Fin-VADER extension and fusion test. The fusion test asked whether the sentiment signal actually improves an economically meaningful portfolio outcome once lagged. It does not, therefore we separate better sentiment measurement from better investment performance.

4. Extensions, Innovation and What the Evidence Changed

This project relies on multiple innovations to deliver an effective product. A new portfolio methodology via HRP and a finance-specific sentiment lexicon are added, as well as a look-ahead safe fusion test, turnover costs, a block bootstrap robustness check, and a design system for FundX. While it's important to note the number of these elements, the focus should be placed on whether they create a measurable effect in my evaluation of the product, its creation or presentation.

Extension	What was implemented	Evidence / decision
HRP	Clustering-based allocation alongside four conventional methods	Adds a distinct, lower-volatility construction rule; does not beat the best benchmark
Fin-VADER	~90 curated finance terms/phrases added to VADER	Neutral share 49.6% -> 45.2%; measurement improves, performance does not
Sentiment fusion	Lagged 21-day sector z-score tilt with bounded multipliers	Sharpe 0.380 -> 0.342; do not market it as alpha
Turnover costs	10 bps x drift-adjusted turnover at rebalances	Penalises Max Sharpe more than Risk Parity; rankings mostly unchanged
Block bootstrap	21-day blocks, 2,000 draws, fixed seed	Sharpe intervals are wide; product decision should not rely on rank alone
Design system	Shared navy/teal/gold visual language across plots and app	Makes evidence and product read as one FundX system

4.1 Fin-VADER: improving the measurement instrument

In addition to supporting a finding in Part A about Plain VADER classifying roughly half of the finance headlines as neutral, the results of Part B confirm that this problem existed prior to examining the extension. Fin-VADER provides a solution for this problem with approximately ninety finance-related word and phrase mapping additions made to Plain VADER's -4 to +4 valence scale. Some examples are: upgrade, downgrade, profit warning, all-time high, bankruptcy, and weak demand. While an artificial intelligence (AI) tool assisted in generating possible term suggestions, the final list of terms along with their corresponding sentiment scores were reviewed and selected by hand as opposed to accepting them without manual review. Additionally, the valence scores were also determined manually rather than estimated from market returns.

Model	Neutral	Mean	Std. dev.	Positive	Negative
Plain VADER	49.6%	+0.106	0.288	37.4%	13.0%
Fin-VADER (extended lexicon)	45.2%	+0.117	0.312	40.6%	14.2%

Figure 7 illustrates that Fin-VADER has altered headline classification. The neutral share dropped from 49.6% to 45.2%. Thus, about 6,500 news stories have been shifted out of the neutral category by the addition of a finance-specific sentiment measure. Additionally, average sentiment rose from 0.106 to 0.117, while the standard deviation increased from 0.288 to 0.312. Also, both the frequency of positive and negative sentiment have grown. That is consistent with its purpose. It identifies clearer direction in financial terms where VADER alone typically identifies a neutral result. However, this does not necessarily mean that the additional sentiment information is useful for trading.

4.2 Fusion: a negative result that changes the product claim

The Fusion Rule was designed with simplicity to enable clarity and reproduction. Therefore, I delay the Sector Sentiment Index by one trading day and then take a 21-day rolling average. After that, I determine the z-score across the ten Sectors to display which sectors exhibit the greatest positive (or negative) relative sentiment. Finally, I adjust each Stock's initial Portfolio Weight utilizing $\text{Clip}(1 + 0.25 Z_{\text{Sector}}, .60, 1.40)$, after which all Weights are normalized again. The one-trading day delay removes Look Ahead Bias; the z-Score represents Relative Sector Sentiment rather than Absolute levels; and the Clipping will prevent the Model from placing too great a Bet in a specific sector. Versions A-D utilize the exact same Equity Maximum Sharpe Model as well as the same Out-of Sample time period utilized in Table 2; therefore this version-to-version comparison within Table 2 is like for like.

Table 2. Fusion comparison OOS 2021-01-04 to 2023-12-29	Ann. ret.	Ann. vol.	Sharpe	Max DD	Turnover	Net Sharpe
Base Equity Max Sharpe	6.6%	17.3%	0.380	-22.6%	28.1%	0.359
+ VADER tilt	6.2%	17.9%	0.344	-22.0%	29.9%	0.323
+ Fin-VADER tilt	6.1%	17.8%	0.342	-21.4%	30.0%	0.320

There was no improvement in the base fund. With basic VADER, Sharpe fell from 0.380 to 0.344, and with Fin-VADER it went down to 0.342; with that same model we have an increase in monthly turnover of 28.1% to approximately 30%. Although there may be a slight reduction in max DD (from -22.6% to -21.4%), Figure 8 demonstrates all three paths were relatively similar and the funds based on the tilted portfolios will end up slightly lower.

The direct prediction analysis was weak. Based on 9800 sector day combinations, the Pearson Information Coefficient was -0.043 and the Spearman Coefficient was -0.023. Both are extremely close to zero and can therefore be considered economically negligible. It's likely that there is no one cause for this result. Sentiment noise exists at a monthly time frame. The Maximum Sharpe Base Fund is already weak out-of-sample. The sentiment indicator is constructed at the sector level while the stocks' weights are established at the individual stock level. Additionally, the 21-day average will smooth out much of the additional variability produced by Fin-VADER. Therefore, Fund X should offer sentiment as an analytical and research tool instead of marketing the tilt as an alpha-generating strategy.

4.3 Turnover, costs and implementation realism

The turnover differences are important as well. Equal Weight moves only about 2.6% to 5.8% of the portfolio each month across the fund families, while Risk Parity turns over about 3.1% to 6.3%. Maximum Sharpe is far more active, with turnover of 28.5% in the combined fund and 28.1% in the equity fund.

At 10 basis points (bps), the overall rankings do not change, but the impact is largest where expected. Combined Maximum Sharpe falls from a Gross Sharpe of 0.744 to a Net Sharpe of 0.727, while Combined Risk Parity only falls from 0.862 to 0.859. Figure 9 also shows that the wealth paths remain similar over the three-year period, meaning the cost assumption is not large enough to drive the result. However, the drag would increase with higher trading costs and remain greatest for the more aggressive optimiser.

4.4 Robustness: the leaderboard is less certain than the point estimates

A 21-day moving-block bootstrap using 2000 draws with random number generator seed 42 alters my perception of the fund ranking. All sampled date blocks are identical across all funds selected in each draw. This means that cross-fund dependency is preserved in comparing pairs of funds. In Figure 10, 90 percent Sharpe intervals are shown. Each interval intersects zero. The Combined Risk Parity model yields a Sharpe ratio of .862 with an estimated interval of approximately [-.25, 2.02] while Combined Maximum Sharpe gives a Sharpe ratio of .744 with an estimated interval of approximately [-.37, 1.61]. The paired Risk Parity minus Max Sharpe difference is +.118 with an estimated interval of approximately [-.41, .95]. Overall, the Risk Parity strategy outperforms the Max Sharpe strategy in 70.9 percent of the draws. Additionally, in 70.6 percent of the paired draws the Fin-VADER sentiment tilt is less effective than its base case. The interval of the paired Fin-VADER strategy also intersects zero. While three years of data are sufficient to demonstrate the existence of drawdowns, turnover differences and the failed fusion result, they do not provide adequate time to determine if these small differences in Sharpe ratios will be persistent over long horizons. Therefore, the FundX recommendations use multiple measurements to determine rankings instead of relying solely on point estimates.

In addition to providing a broader range of portfolio strategies through HRP, testing implementation drag due to transaction costs, evaluating the ability of sentiment measurement to be enhanced through Fin-VADER and testing whether such enhancements translate into better portfolio performance through fusion, the nearly-zero information coefficient also indicates that there may be limited predictive value associated with the signal itself at this horizon. Finally, the bootstrap allows for determining whether the ranking of funds is robust. Thus, the order in which the extensions occur is important because an extension could successfully function at one level and then fail at another. For example, Fin-VADER was successful in improving sentiment measurement however was unsuccessful in enhancing Sharpe ratios. Similarly, the bootstrap prevented small differences in point estimates from being perceived as certainties. As a result, I consider the negative fusion result to be an innovation outcome because it provides evidence regarding what occurred, what did not occur and how such results should impact the FundX product claims.

5. The App and the Investor Journey

The user will have access to the same content from the Streamlit App that was previously used for the analysis. However, in this case, it has been turned into an Investor Journey with 7 different pages. The pages are ordered sequentially so users may begin by learning about all of the available investment funds, then compare one or multiple funds based on their characteristics, and ultimately build their own investment portfolio. Users can also access the methodology and data behind the resulting allocations, meaning the process remains transparent rather than operating as a black box.

Journey step	What the user can do	Why it matters
1. Discover	Overview + Find my mix introduce the shelf and risk preferences	Reduces a 17-fund universe to a manageable starting point
2. Compare	Filter funds, sort metrics and overlay growth of \$1	Makes return, risk and drawdown trade-offs visible side by side
3. Inspect	Open a fact sheet with KPIs, drawdown, holdings and weight history	Turns each rule into an explainable investable product
4. Allocate	Combine funds, normalise to 100%, enter amount/horizon and view fee illustration	Connects the backtest to an actual portfolio decision
5. Investigate	Switch VADER/Fin-VADER, change smoothing and inspect methodology/coverage	Keeps sentiment and model limitations visible rather than hidden

Beginning with the Allocation Builder, it gives shape to the Part A Business Model. In allocating equally (one third) to each of the three strategies (Combined Maximum Sharpe, Combined Risk Parity & Equity Minimum Variance), the strategy yields approximately 12.6 percent annualized return, 15.6 percent volatility, .81 Sharpe ratio and -18.9 percent max. draw down across the common out-of-sample (OOS) history. Using an example of a \$10,000 investment, the annualized 0.75 percent management fee would be approximately \$75 per year. As these are historical (not predictive) results; the main purpose of this feature is to show users the actual result of their allocation choice.

There was no need for heavy development to create the current application. All calculations for optimization were done outside of StreamLit by the scripts/run_part_b.py script. Only pre-computed file results located within results/, are read into memory using streamlit. No usage of Vader or NLTK, nor any optimizer occur at runtime. It allowed for a minimalistic approach to maintain responsiveness when deployed to the free tier service provider. The report and web application share the same FundX Navy Blue / Teal / Gold design theme which provides a sense of continuity between the presentation of the analytical evidence and the User Interface of the product.

The web application provides additional functionality as a "control" for communicating the outcome of the analysis to the user. Users do not view simply a leaderboard of funds ranked by their Sharpe ratios. Instead, users can examine return, volatility, drawdown and compound growth simultaneously via the Comparison Page. Additionally, users have access to examine holdings and the evolution of Target Weights via the Fact Sheet. For the Combined Maximum Sharpe portfolio, there is considerable concentration among the five largest final positions. Together they represent 78.4 percent of the Target Portfolio even though there is a limit of 20 percent per position. With this knowledge, users may now ask themselves what combination of funds will best fit their level of risk tolerance and investment size?

The Sentiment Analytics area has been positioned beside Coverage & Methodology rather than being displayed as a buy/sell signal. This is consistent with our finding that Sentiment Analytics can provide value as an Analytic tool, however we have not found sufficient evidence to support its use as Alpha. Therefore, the Product Design follows the Evidence Hierarchy in the Report: users can observe historical performance, review assumptions made and view experimental features without implying them to be anything other than experimentation. Transparency is crucial in supporting the needs of our target user. He/she desires assistance with constructing a portfolio without transferring all responsibility for making decisions to an unknown Black Box.

LIVE PRODUCT

App: <https://z5544937-fundx.streamlit.app> | Public repo: https://github.com/1Tehan/z5544937_projectB

6. Critical Reflection and Three Recommendations

The first important lesson I took away from my experience with FundX is that the simplest strategy doesn't have to be the worst in terms of the quality of your final product. I assumed that optimization strategies would continue to dominate the simple benchmark as soon as we moved to a walk-forward backtesting process, however that did not happen. All optimized equity funds were beaten by the "Equity Equal Weight" on Sharpe ratio. Additionally, all Maximum Sharpe funds used had clear concentration problems along with high turnover and relied on unstable expected-return estimates. For this reason, I determined that Risk Parity would be the lead for FundX. Rather than seeking to create an optimal allocation based on past returns, Risk Parity seeks to get each asset contributing equally to total portfolio risk. And based on this example, the less complicated concept performed better out of sample.

I also found that simply improving measurement does not guarantee you'll achieve better investment performance. Even though Fin-VADER improved the sentiment measurements, the neutral share fell by 4.4 percentage points and the scores are now more directional; the performance of the portfolio didn't increase. And since the Information Coefficient is nearly zero, it appears that the signal has almost no predictive ability

at this time frame. As such, even though the study produced a negative result, it is still positive for FundX, as it prevents FundX from claiming performance based on a sentiment feature with limited predictive power.

Similarly, the results of the cryptocurrency testing demonstrated another way that a retail product cannot be assessed solely upon return. At first glance, a 59.8% return per annum sounds like a tremendous performance metric; however, when compared to the -72.0% drawdown, it produces a very different picture. Therefore, I believe there is stronger evidence to support the notion of utilizing cryptocurrency within risk-budgeting or combination portfolios rather than using a crypto-only fund as the default.

There are many obvious constraints as well. First and foremost, there are only three years of out-of-sample data. Secondly, the cost model is very basic. Specifically, I am assuming a flat 10 bps transaction-cost assumption and excluding spread costs, market impact costs and other fees associated with distressed market conditions. Thirdly, I utilize headline news only and not full articles for my sentiment analysis, and I set the Fin-VADER valences manually instead of statistically estimating them. While these constraints don't eliminate the usefulness of my findings, they do demonstrate what needs to be examined next.

Therefore, my suggestions are directly related to those constraints. Data quality should be a priority because one poor return may negatively affect the estimation of covariances for every fund developed from those returns. The second recommendation would be to determine if the hand-set valences could be replaced with actual evidence derived from returns. The third recommendation would be to implement turnover controls because they directly address one of the major shortfalls of Maximum Sharpe and can be analyzed through the lens of gross vs net metrics. Finally, I wouldn't prioritize adding additional optimization techniques prior to addressing these items. My experience demonstrates that simply having additional options for optimizing does not inherently provide a better OOS product. Both Equal Weight and Risk Parity are highly competitive.

Recommendation 1 - Productionise the Part A extreme-move screen

In Part A, the first version of this code retained 29 observations where $|z| > 8$. To get more out of this work, the second version will transform this screen into a real-time quality-control monitoring tool for prices, which could flag extreme moves as soon as they occur so they do not influence returns, covariance estimates or portfolio weights.

Recommendation 2 - Estimate, rather than hand-set, finance valences

FIN-VADER proves that financial vocabulary changes classification; however, the existing valences remain curated judgments, and the almost zero Information Coefficient indicates very little predictive content from the signal at this horizon.

The Next Version of FIN-VADER will be to run an Ordinary Least Squares (OLS) regression model, using abnormal returns from the next trading day as the dependent variable and dummy variables for each term present from the roughly 90-word lexicon. I will estimate the model using data from 2020 through 2022 and then evaluate it on a held-out sample of 2023 data. I would retain only those terms whose direction and statistical evidence survive out-of-sample testing. Then I would rerun the exact same Fusion Rule without making any adjustments or retuning it.

Recommendation 3 - Put turnover inside the optimiser

Combined Maximum Sharpe has monthly portfolio turnover of about 28.5%, whereas Combined Risk Parity is approximately 3.7%. In a commercial implementation it would be advisable to include either a trading cost (turnover) penalty, or no-trade bands around existing positions, and therefore optimise net-of-cost returns.

FINAL FUNDX POSITION

Balanced growth – Combined Risk Parity

Defensive users – Minimum Variance

Equal Weight remains the benchmark. Crypto-only funds remain high-risk, and sentiment should be presented as an analytic until a stronger fusion test shows incremental value.

References

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Appendix A - Required Exhibits and Supporting Evidence

The appendix contains the required Part B exhibits plus the block-bootstrap robustness figure. Every exhibit is generated from the committed real-data results and is referenced and interpreted in the written narrative. The full performance table includes net Sharpe as an additional implementation metric.

Fund	Ann. ret.	Ann. vol.	Sharpe	Max DD	Growth \$1	Turnover	Net Sharpe
Combined Equal Weight	15.1%	21.6%	0.701	-27.9%	\$1.52	3.8%	0.698
Combined Minimum Variance	5.5%	12.6%	0.434	-15.5%	\$1.17	13.0%	0.421
Combined Maximum Sharpe	17.3%	23.2%	0.744	-22.7%	\$1.61	28.5%	0.727
Combined Risk Parity (ERC)	14.0%	16.2%	0.862	-19.5%	\$1.48	3.7%	0.859
Combined Hierarchical Risk Parity	9.8%	14.0%	0.701	-17.9%	\$1.32	10.3%	0.691
Equity-only Equal Weight	12.6%	16.1%	0.784	-20.2%	\$1.43	2.6%	0.782
Equity-only Minimum Variance	5.5%	12.6%	0.436	-15.3%	\$1.17	12.8%	0.423
Equity-only Maximum Sharpe	6.6%	17.3%	0.380	-22.6%	\$1.21	28.1%	0.359
Equity-only Risk Parity (ERC)	9.9%	14.5%	0.682	-18.5%	\$1.33	3.1%	0.679
Equity-only Hierarchical Risk Parity	8.4%	13.7%	0.611	-17.4%	\$1.27	11.9%	0.600
Crypto-only Equal Weight	40.5%	81.9%	0.495	-81.6%	\$2.77	5.8%	0.493
Crypto-only Minimum Variance	59.8%	71.6%	0.836	-72.0%	\$4.08	10.2%	0.833
Crypto-only Maximum Sharpe	-3.0%	76.0%	-0.040	-85.6%	\$0.91	21.8%	-0.043
Crypto-only Risk Parity (ERC)	44.3%	79.9%	0.555	-79.9%	\$3.01	6.3%	0.553
Crypto-only Hierarchical Risk Parity	43.2%	76.9%	0.562	-78.1%	\$2.94	10.4%	0.559
Equity Max Sharpe + Sentiment Tilt (VADER)	6.2%	17.9%	0.344	-22.0%	\$1.20	29.9%	0.323
Equity Max Sharpe + Sentiment Tilt (Fin-VADER)	6.1%	17.8%	0.342	-21.4%	\$1.19	30.0%	0.320

Table 1. Fact-sheet performance metrics for all 17 funds. Equity and combined funds: OOS 2021-01-04 to 2023-12-29, annualisation x252. Crypto-only funds: OOS 2021-01-01 to 2023-12-31, annualisation x365. $r_f = 0$. Net Sharpe uses 10 bps x turnover.

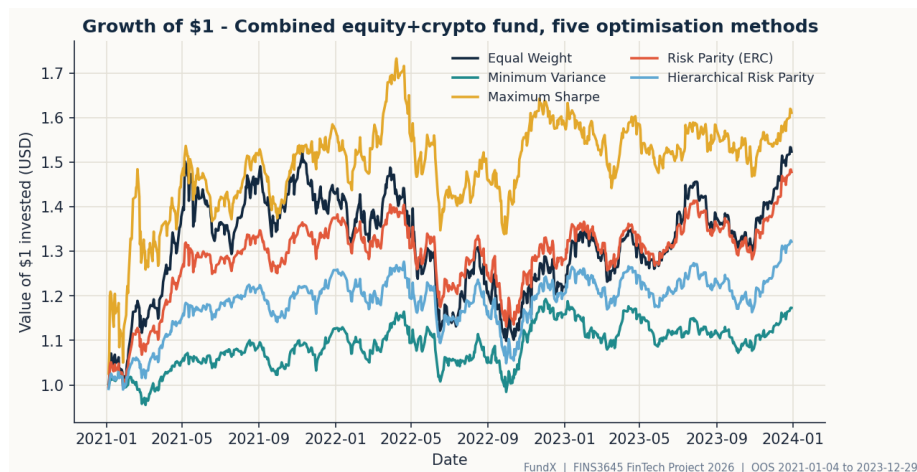


Figure 1. Growth of \$1 for the combined equity+crypto family under five allocation methods. Units: USD value of \$1 invested; OOS 2021-01-04 to 2023-12-29.

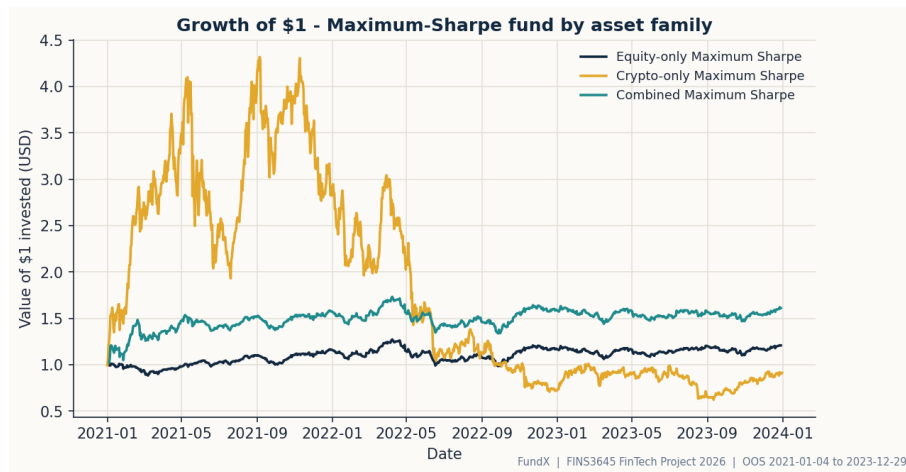


Figure 2. Growth of \$1 for Maximum-Sharpe funds by asset family. Units: USD value of \$1 invested; equity/combined OOS 2021-01-04 to 2023-12-29, crypto-only OOS 2021-01-01 to 2023-12-31.

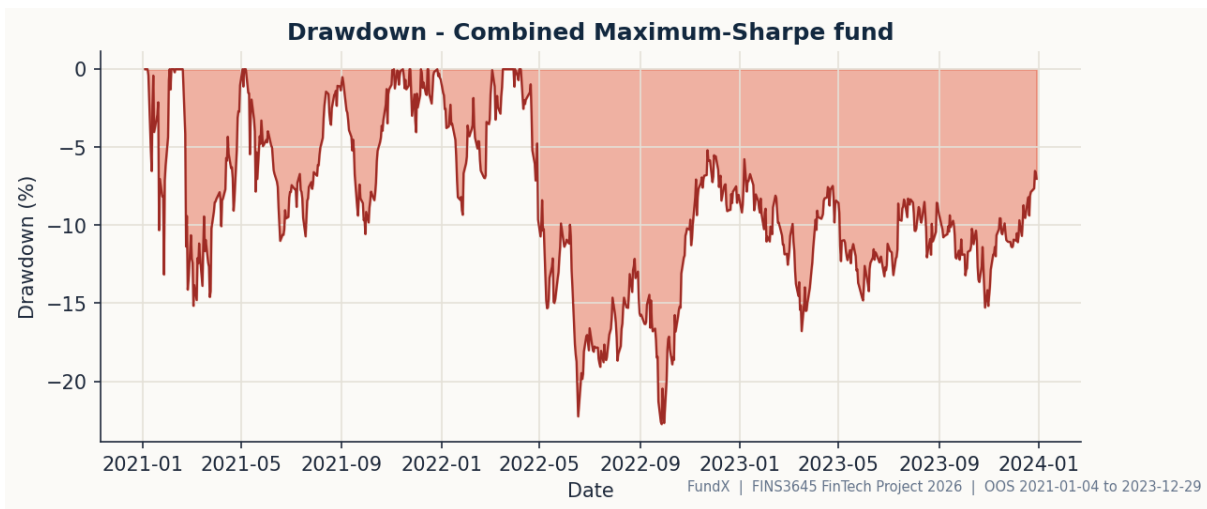


Figure 3. Drawdown from the prior wealth peak for Combined Maximum Sharpe. Units: percentage below prior peak; OOS 2021-01-04 to 2023-12-29.

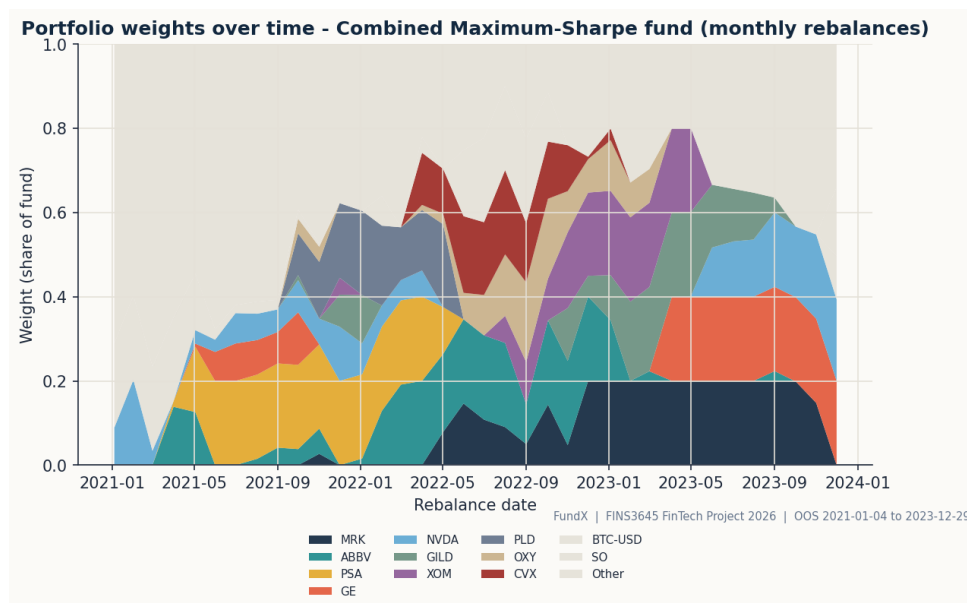


Figure 4. Target weights at monthly rebalances for Combined Maximum Sharpe. Units: share of fund; explicit 20% per-asset cap; OOS rebalances 2021-01-04 to 2023-12-01.

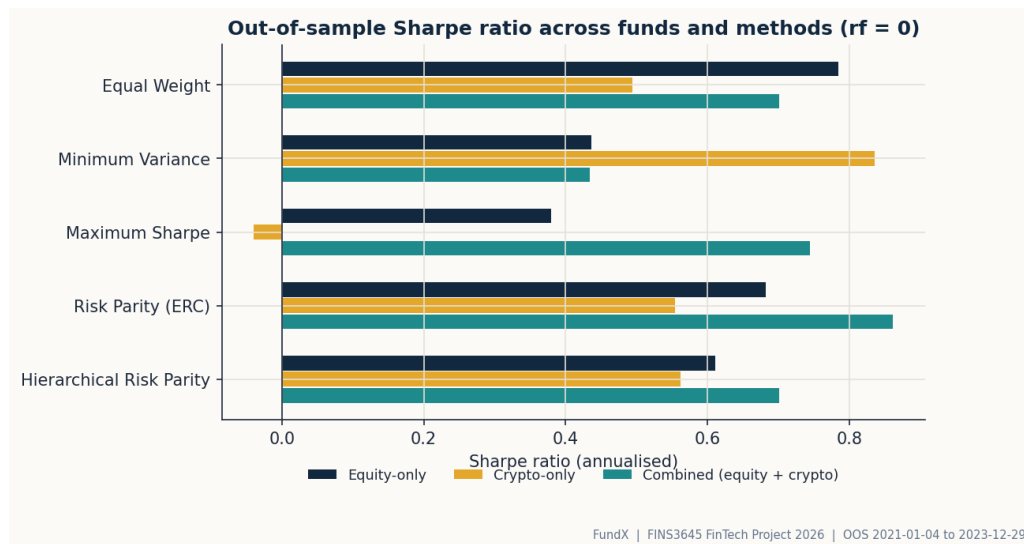


Figure 5. Annualised out-of-sample Sharpe ratios across fund families and allocation methods, $r_f = 0$. Equity/combined OOS 2021-01-04 to 2023-12-29; crypto-only OOS 2021-01-01 to 2023-12-31.

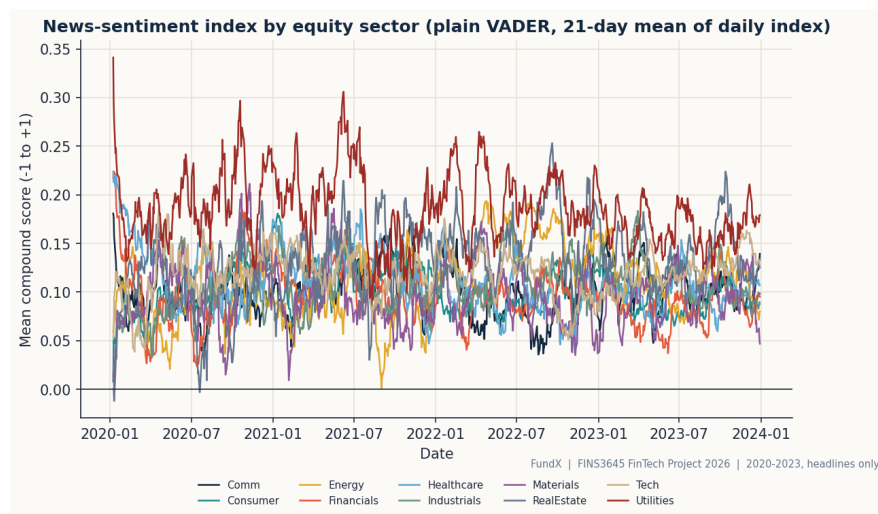


Figure 6. Equity-sector news-sentiment index: 21-day mean of the daily plain-VADER sector index. Units: mean compound score (-1 to +1), 2020-2023 headlines only.

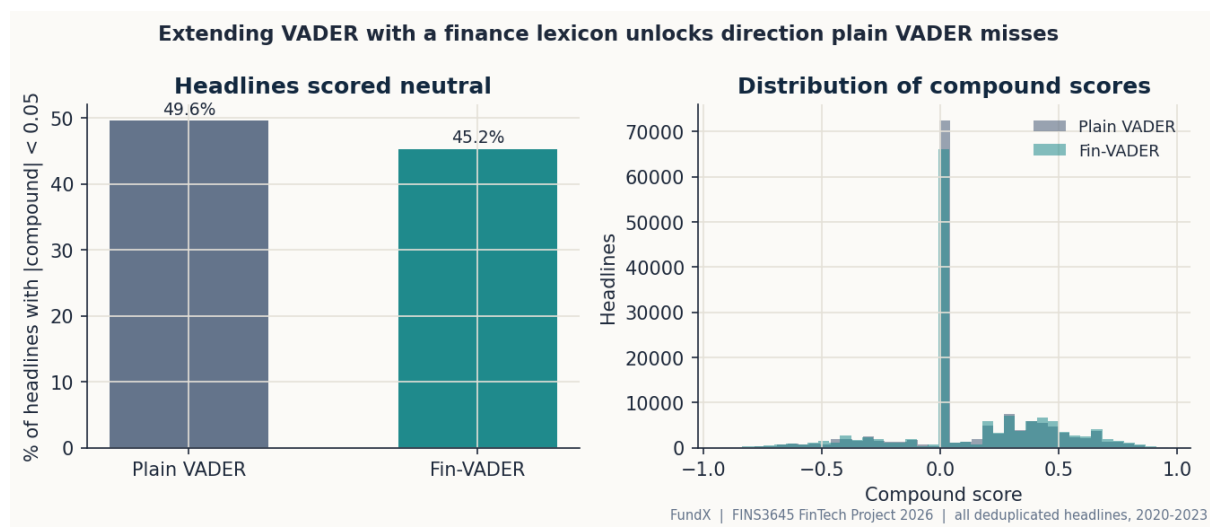


Figure 7. Fin-VADER measurement effect for all 146,836 deduplicated 2020-2023 headlines. Left: neutral share (%). Right: compound-score distribution (-1 to +1).

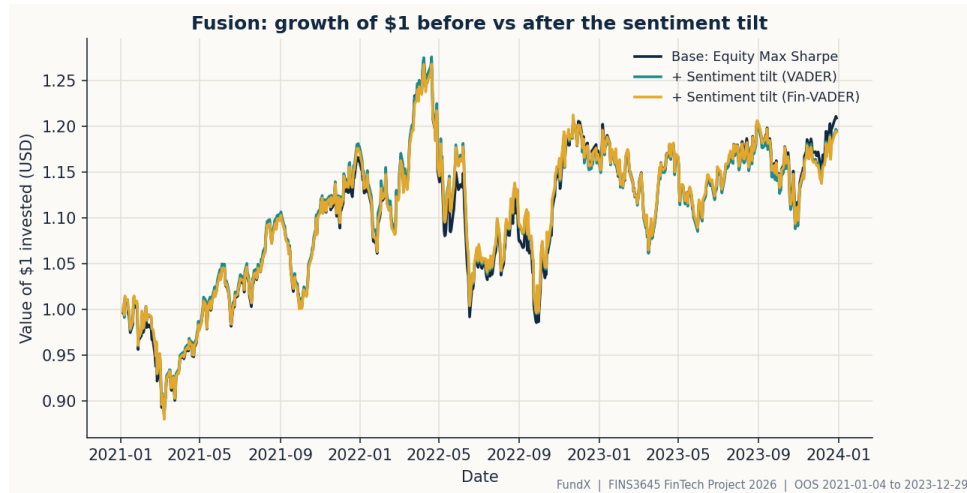


Figure 8. Growth of \$1 before and after the sentiment tilt: Equity Maximum Sharpe base versus VADER and Fin-VADER variants. Units: USD value of \$1 invested; OOS 2021-01-04 to 2023-12-29.

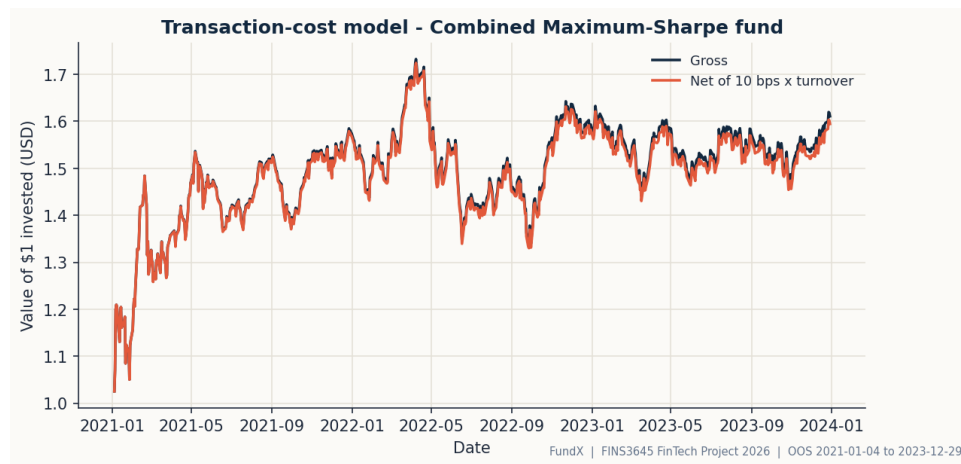


Figure 9. Transaction-cost model for Combined Maximum Sharpe: gross versus net-of-cost growth of \$1 using 10 bps x turnover. Units: USD value of \$1 invested; OOS 2021-01-04 to 2023-12-29.

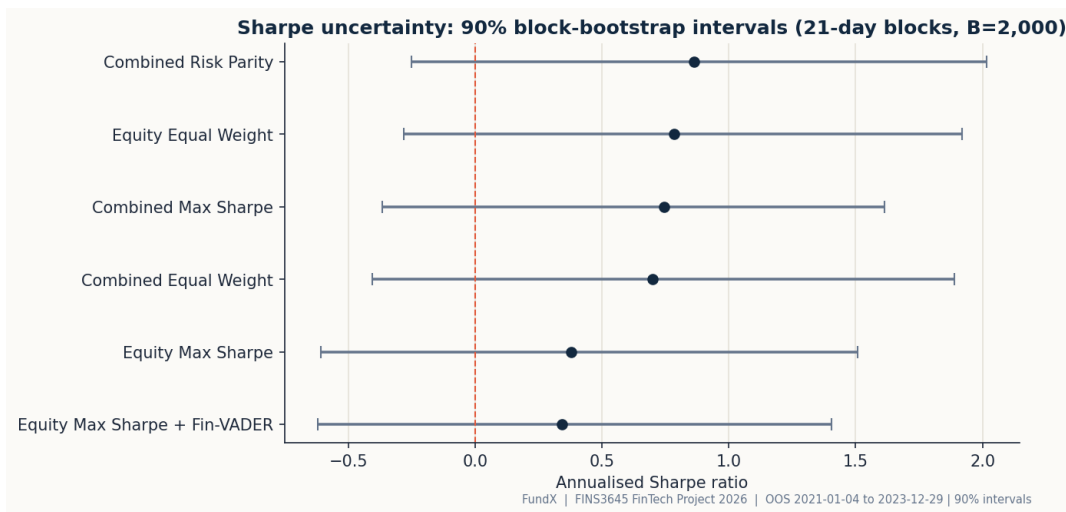


Figure 10. Annualised Sharpe ratios ($r_f = 0$) with 90% moving-block-bootstrap intervals for selected funds; 21-day blocks, 2,000 draws, seed 42. OOS 2021-01-04 to 2023-12-29. Every displayed interval spans zero.